

Physics 3130H: Worksheet

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Problem 1. (a) Function representing curl is a _____

(b) Function representing gradient is a _____

(c) gravitational field is a _____

(d) gravitational potential is a _____

(e) forces are _____

(f) force component is a _____

(g) Speed is a _____

(h) velocity is a _____

(i) angular velocity is a _____

(j) angular acceleration is a _____

(k) potential energy is a _____

(l) kinetic energy is a _____

Solution 1.

(a) Function representing curl is a **vector**.

(b) Function representing gradient is a **vector**.

(c) gravitational field is a **vector**.

(d) gravitational potential is a **scalar**.

(e) forces are **vectors**.

(f) force component is a **scalar**.

(g) speed is a **scalar**.

(h) velocity is a **vector**.

(i) angular velocity is a **vector**.

(j) angular acceleration is a **vector**.

(k) potential energy is a **scalar**.

(l) kinetic energy is a **scalar**.

Problem 2. An object moves in the xy -plane in such a way that its position vector $\mathbf{r}$ is given as a function of time t by $\mathbf{r} = a \cos \omega t \hat{i} + b \sin \omega t \hat{j}$ where a , b , and ω are constants.

- (a) How far is the object from the origin at any time t ?
- (b) Find the object's velocity and acceleration as a function of time.
- (c) Show that the object moves on the elliptical path.

Solution 2.

- (a) If we take the origin to be $\mathbf{O} = (0, 0, 0)$ then its distance from the origin is $r = \sqrt{a^2 \cos^2 \omega t + b^2 \sin^2 \omega t}$.
- (b) $\mathbf{v} = d\mathbf{r}/dt = -a\omega \sin \omega t \hat{i} + b\omega \cos \omega t \hat{j}$. $\mathbf{a} = d\mathbf{v}/dt = -a\omega^2 \cos \omega t \hat{i} - b\omega^2 \sin \omega t \hat{j}$.
- (c) It is clear enough to see the elliptic motion which arises from these equations. The object's position (r , distance from the origin) is a strictly oscillating function, i.e. the object is confined to orbit around the origin.

Problem 3. Calculate the divergence of each of the following functions:

- (a) $x^2 \hat{i} + y^2 \hat{j} + z^2 \hat{k}$.
- (b) $zy \hat{i} + xz \hat{j} + xy \hat{k}$.
- (c) $e^{-x} \hat{i} + e^{-y} \hat{j} + e^{-z} \hat{k}$.

Solution 3.

- (a) $\nabla \cdot (x^2 \hat{i} + y^2 \hat{j} + z^2 \hat{k}) = 2x + 2y + 2z$.
- (b) $\nabla \cdot (zy \hat{i} + xz \hat{j} + xy \hat{k}) = 0$.
- (c) $\nabla \cdot (e^{-x} \hat{i} + e^{-y} \hat{j} + e^{-z} \hat{k}) = -e^{-x} - e^{-y} - e^{-z}$.

Problem 4. Calculate the curl of each of the following functions:

- (a) $xy \hat{i} + yz \hat{j} - y^2 \hat{k}$.
- (b) $xe^{-x} \hat{i} + e^{-y} \hat{j} + e^{-z} \hat{k}$.
- (c) $x \hat{i} + y \hat{j} + (x^2 + y^2) \hat{k}$.

Solution 4.

- (a) $\nabla \times (xy \hat{i} + yz \hat{j} - y^2 \hat{k}) = [-2y - y] \hat{i} + 0 \hat{j} + x \hat{k}$.
- (b) $\nabla \times (xe^{-x} \hat{i} + e^{-y} \hat{j} + e^{-z} \hat{k}) = \mathbf{0}$.
- (c) $\nabla \times (x \hat{i} + y \hat{j} + (x^2 + y^2) \hat{k}) = 2y \hat{i} + 2x \hat{j} + 0 \hat{k}$.

Problem 5. For any central force show that curl of $\mathbf{F}$ is zero. (use spherical polar coordinates: Refer to the lecture slides in gravitation chapter and central forces chapter)

Solution 5. Central forces are forces of the form $\mathbf{F}(\mathbf{r}) = F(r)\hat{\mathbf{r}}$. Curl in spherical coordinates is given by

$$\begin{aligned} \nabla \times \mathbf{v} &= \frac{1}{r \sin \theta} [\partial_\theta (\sin \theta v_\phi) - \partial_\phi v_\theta] \hat{\mathbf{r}} \\ &+ \frac{1}{r} \left[\frac{1}{\sin \theta} \partial_\phi v_r - \partial_r (rv_\phi) \right] \hat{\boldsymbol{\theta}} \\ &+ \frac{1}{r} [\partial_r (rv_\theta) - \partial_\theta v_r] \hat{\boldsymbol{\phi}}. \end{aligned}$$

Clearly since the curl does not contain a term like $\partial_r v_r$ it will be zero. Note that under this definition a central force is one which is spherically symmetric. A nonsymmetric central force which looks like $\mathbf{F}(\mathbf{r}) = F(r, \theta, \phi)\hat{\mathbf{r}}$ may not have vanishing curl.

Problem 6. Objective is to find the gravitational field. For the following configurations write down the corresponding integrals:

- (a) A rod of length L (at a point R along the length ($R > L/2$) from the center, at a point R perpendicular to the length from the center of the rod)
- (b) A spherical shell (at a point inside the shell, in between the inner and outer radius, at a point outside the sphere)
- (c) A hollow sphere (at a point inside the shell, and outside the shell)
- (d) A solid sphere (at a point inside the shell, and outside the shell)

Solution 6. Note that for the most part here I am just looking at setting up the integrals. I'm not going to make simplifications.

- (a) The general form for gravitational field is (using the notational conventions from Griffiths' electrodynamics)

$$\mathbf{g} = -G \int \frac{\rho(\mathbf{r}') \hat{\mathbf{z}}}{r^2} d^3 r'.$$

Here we ought to take advantage of cylindrical symmetry and integrate over s, ϕ, z . Additionally, we take this rod, given that we are not told a radius, to have a linear mass density $\lambda(\mathbf{r}')$. If we are at a point R along the length of the rod then $\mathbf{r} = 0\hat{\mathbf{s}} + 0\hat{\phi} + R\hat{\mathbf{z}}$ and $\mathbf{r}' = z'\hat{\mathbf{z}}$ so $\mathbf{z} = R - z'\hat{\mathbf{z}}$ and the integral becomes

$$\mathbf{g} = -G \int_0^L \frac{\lambda(z') [R - z']}{|R - z'|^3} dz' \hat{\mathbf{z}}.$$

When we are at a distance R from the center of the rod along $\hat{\mathbf{s}}$ we have that $\mathbf{r} = R\hat{\mathbf{s}}$ and $\mathbf{r}'$ is unchanged. This gives $\mathbf{z} = R\hat{\mathbf{s}} - z'\hat{\mathbf{z}}$ and so

$$\mathbf{g} = -G \int_0^L \frac{\lambda(z') [R\hat{\mathbf{s}} - z'\hat{\mathbf{z}}]}{[R^2 + (z')^2]^{3/2}} dz'.$$

- (b) Inside the shell (of volume mass density $\rho(\mathbf{r}')$), $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, $\mathbf{r}' = x'\hat{x} + y'\hat{y} + z'\hat{z} = r' \sin \theta' \cos \phi' \hat{x} + r' \sin \theta' \sin \phi' \hat{y} + r' \cos \theta' \hat{z}$ and so

$$\mathbf{z} = (x - r' \sin \theta' \cos \phi') \hat{x} + (y - r' \sin \theta' \sin \phi') \hat{y} + (z - r' \cos \theta') \hat{z}.$$

The integral then becomes

$$\mathbf{g} = -G \int_0^{2\pi} \int_0^\pi \int_0^R \frac{\rho(\mathbf{r}') [(x - r' \sin \theta' \cos \phi') \hat{x} + (y - r' \sin \theta' \sin \phi') \hat{y} + (z - r' \cos \theta') \hat{z}]}{[(x - r' \sin \theta' \cos \phi')^2 + (y - r' \sin \theta' \sin \phi')^2 + (z - r' \cos \theta')^2]^{3/2}} (r')^2 \sin \theta' dr' d\theta' d\phi'$$

where I'm assuming the mass density will account for the hollow portion. If this is not the case the bounds on the r' integral could be adjusted to integrate only over the real mass. Between the inner and outer shell the integral remains the same, the values of x, y, z would just change. The same is the case for outside the sphere.

- (c) I'll skimp on the details because I think I get the point. The integral is

$$\mathbf{g} = -G \int_0^{2\pi} \int_0^\pi \frac{\sigma(\mathbf{r}') [(x - R \sin \theta' \cos \phi') \hat{x} + (y - R \sin \theta' \sin \phi') \hat{y} + (z - R \cos \theta') \hat{z}]}{[(x - R \sin \theta' \cos \phi')^2 + (y - R \sin \theta' \sin \phi')^2 + (z - R \cos \theta')^2]^{3/2}} (R)^2 \sin \theta' d\theta' d\phi'$$

where I've assumed the shell has surface mass density σ and radius R . Again this integral is the same inside and out, the values of x, y, z will just change.

- (d) This is the same integral as in part (b), just with $\rho(\mathbf{r}')$ nonzero for the entire sphere. In the case we would not have to change the bounds on the r' integral as we are integrating over the entire spherical volume.

Problem 7. Calculate the Coriolis force for an object falling vertically down in the southern hemisphere. I am interested to see the components of the vector quantities involved, the vector product, and the direction of the lateral deflection.

Solution 7. The Coriolis force is given by $-2m\boldsymbol{\omega} \times \mathbf{v}_r$ where $\boldsymbol{\omega}$ is the angular velocity of the rotating body which causes the force and $\mathbf{v}_r$ is the velocity of the falling object as measured in the rotating frame, which here is the surface of the Earth. Recall that in class we defined $\boldsymbol{\omega} = -\omega \cos \lambda \hat{x} + \omega \sin \lambda \hat{z}$ with $\hat{x}$ pointing south, $\hat{y}$ pointing east, and $\hat{z}$ normal to the Earth's surface. ω is the angular *speed* and λ is the latitude, running from $+\pi/2$ at the north pole to 0 at the equator and $-\pi/2$ at the south pole. Since we are in the southern hemisphere, $|\lambda| = -\lambda$ so $\boldsymbol{\omega} = -\omega \cos \lambda \hat{x} - \omega \sin |\lambda| \hat{z}$. Assuming the object starts from rest and begins falling (e.g. it wasn't thrown upwards, or if it was we start our measurement from the throw's apex) then $\mathbf{v}_r = -gt\hat{z}$ if we take gravity to be along $\hat{z}$ alone which is not quite correct but is sufficient. This then gives a cross product of the form

$$-\boldsymbol{\omega} \times \mathbf{v}_r = - \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ -\omega \cos \lambda & 0 & -\omega \sin |\lambda| \\ 0 & 0 & -gt \end{vmatrix} = 0\hat{x} + gt\omega \cos \lambda \hat{y} + 0\hat{z}$$

so the deflection is entirely along $\hat{y}$ and points east.